

European Options

1 What is an Option?

An option is a financial derivative — a contract whose value is derived from the price of some underlying asset. The underlying can be almost anything: an equity, a stock index, a foreign currency, a commodity, even an interest rate.

What makes options distinctive is the asymmetry they create. Unlike buying a stock outright, buying an option gives you the *right*, but not the *obligation*, to transact at a pre-agreed price. You pay a premium upfront for this right. If it turns out the option would cost you money to exercise, you simply don't.

There are two fundamental types:

- A **call option** gives the holder the right to *buy* the underlying asset at the agreed price.
- A **put option** gives the holder the right to *sell* the underlying asset at the agreed price.

The pre-agreed price is called the **strike** (or exercise price), denoted K . The date on which the right expires is the **expiry** (or maturity), denoted T (measured in years from today).

2 What Makes an Option “European”?

The term *European* refers to the **exercise style**, not the geography. A European option can only be exercised on its expiry date — not before. If you hold a European call and the underlying is trading well above the strike on day 180 of a 365-day contract, you cannot lock in that profit yet; you have to wait until day 365.

This is in contrast to an **American option**, which can be exercised at any point up to and including expiry. The added flexibility of American options makes them at least as valuable as their European counterparts, and the early-exercise feature complicates pricing considerably — there is no general closed-form solution. The Black-Scholes formula applies to European options; pricing American options typically requires numerical methods.

3 Payoff at Expiry

The payoff of an option at expiry depends on the relationship between the final spot price S_T and the strike K .

For a **European call**:

$$\text{Payoff}_{\text{call}} = \max(S_T - K, 0)$$

If the stock finishes above the strike, the holder exercises and profits $S_T - K$. If it finishes below, the option expires worthless and the payoff is zero — there is no reason to buy at K when you can buy in the market for less.

For a **European put**:

$$\text{Payoff}_{\text{put}} = \max(K - S_T, 0)$$

If the stock finishes below the strike, the holder exercises and profits $K - S_T$ (selling at K when the market price is lower). If it finishes above, the option expires worthless.

The $\max(\cdot, 0)$ structure is precisely what creates the asymmetry: losses are capped at the premium paid, while gains are theoretically unlimited (for calls) or bounded by K (for puts, since a stock price cannot fall below zero).

4 Moneyness

The relationship between the current spot price S and the strike K is described by the option's *moneyness*:

Moneyness	Call condition	Put condition
In-the-money (ITM)	$S > K$	$S < K$
At-the-money (ATM)	$S \approx K$	$S \approx K$
Out-of-the-money (OTM)	$S < K$	$S > K$

An ITM option has positive *intrinsic value* — what it would be worth if exercised immediately. An OTM option has zero intrinsic value, but may still have positive *time value* because there is a chance the underlying moves favourably before expiry.

The option's total price (premium) is:

$$\text{Premium} = \frac{\text{Intrinsic value}}{\max(S-K, 0) \text{ for call}} + \text{Time value}$$

5 The EuropeanOption Class

The `EuropeanOption` dataclass represents the contract specification — it captures the terms agreed at inception, independent of what the market is doing today.

```
from options_pricer import EuropeanOption, OptionType

# A call option with strike 105, expiring in 6 months (0.5 years)
call = EuropeanOption(option_type=OptionType.CALL, strike=105.0, expiry=0.5)

# A put option with strike 95, expiring in 1 year
put = EuropeanOption(option_type=OptionType.PUT, strike=95.0, expiry=1.0)

# OptionType can also be passed as a plain string
call_alt = EuropeanOption(option_type="call", strike=105.0, expiry=0.5)
```

The constructor validates both fields on creation:

```
# These will both raise ValueError
EuropeanOption(option_type=OptionType.CALL, strike=-10.0, expiry=1.0) # negative strike
EuropeanOption(option_type=OptionType.CALL, strike=100.0, expiry=0.0) # zero expiry
```

`EuropeanOption` represents the *contract* only — it says nothing about what the underlying is currently trading at, what interest rates are, or what volatility looks like. Those inputs live in `MarketData`. The separation means you can reprice the same contract against many different market scenarios without recreating it.

```
from options_pricer import BlackScholes, MarketData

contract = EuropeanOption(option_type=OptionType.CALL, strike=100.0, expiry=1.0)

# Price under baseline market conditions
md_base = MarketData(spot=100.0, rate=0.05, vol=0.20)
print(BlackScholes.price(contract, md_base))

# Price if volatility spikes to 40%
md_high_vol = MarketData(spot=100.0, rate=0.05, vol=0.40)
print(BlackScholes.price(contract, md_high_vol))
```